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Chapter - 1
System of Linear Equation

1. Define Linear, homogeneous and non-homogeneous. [2019]
2. Define degenerate and non-degenerate linear equations. [2019, 2018]
3. Define consistent.
4. Define free variable.
5. Reduce the following system of linear equation to an echelon system and find the solution:-

$$2x + y - 2z = 10$$
$$3x + 2y + 2z = 1$$
$$5x + 4y + 3z = 4$$

[2018]

6. Find the non-trivial solution for the system of linear equations-

$$x + 2y - 3z = 0$$
$$2x + 5y + 2z = 0$$
$$3x - y - 4z = 0$$

[2018]

7. Solve the following system-

$$x - 2y - 3z = 0$$
$$-x + 2y - 3z = 0$$
$$x + y + 4z = 0$$

8. Find the general solution and also a particular solution of the following system of linear equations:-

$$2x_1 - x_2 - x_3 + 3x_4 = 1$$
$$4x_1 - 2x_2 - x_3 + x_4 = 5$$
$$6x_1 - 3x_2 - x_3 - x_4 = 9$$
$$2x_1 - x_2 + 2x_3 - 12x_4 = 10$$

9. Express the following system of linear equations to echelon form and solve it:-

$$x + y = -3$$
$$2x - 2y - z = -8$$
$$4x - z = -14$$
$$x - 3y - z = -5$$

10. Reduce the following system of linear equations to an echelon form and find all Solutions:-

[2020]

$$x_1 - x_2 + 2x_3 = 5$$
$$2x_1 + x_2 - x_3 = 2$$
$$2x_1 - x_2 - x_3 = 4$$
$$x_1 + 3x_2 + 2x_3 = 1$$

11. Determine the values of λ such that the following system of linear equations in unknowns x, y and z has i) a unique solution, ii) no solution, iii) more than one solution:

$$x + y + \lambda z = 1$$
$$x + \lambda y + z = \lambda$$
$$\lambda x + y + z = \lambda^2$$

[2020, 2018]

12. Determine the value of λ such that the following system in unknowns x, y and z has (i) unique solution; (ii) no solution, (iii) more than one solution-

$$x + y - z = 1$$
$$2x + 3y + \lambda z = 3$$
$$x + \lambda y + 3z = 2$$

13. For What values of λ and μ the following system of linear equations has (i) no solution (ii) more than one solution (iii) a unique solution:-

$$x + y + z = 6$$
$$x + 2y + 3z = 10$$
$$x + 2y + \lambda z = \mu$$

14. Determine the values of λ such that the following system in unknowns x, y and z (i) a unique solution, (ii) no solution, (iii) more than one solution:-

$$\lambda x + y + z = 1$$
$$x + \lambda y + z = 1$$
$$x + y + \lambda z = 1$$

15. Find out the conditions on α, β and γ so that the following systems of non-homogeneous linear equations has a solution:-

$$x + 2y - 3z = \alpha$$
$$2x + 6y - 11z = \beta$$
$$2x - 4y + 14z = 2\gamma$$

Chapter - 2
Determinants

1. Define Determinant.

2. Define minors and co-factors. [2019]

3. Write down the properties of determinants.

4. Define adjoint determinant.

5. Define inverse or reciprocal determinant.

6. Define symmetric and skew-symmetric determinant.

7. Define ortho-symmetric determinant

8. Explain the expansion of determinant

Or, Explain the Laplace expansion.

9. If 2s = a + b+ c prove that-

$$\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix}$$ [2020]

10. . Prove that-

$$\begin{vmatrix} 1+a_1 & a_2 & a_3 & a_4 \\ a_1 & 1+a_2 & a_3 & a_4 \\ a_1 & a_2 & 1+a_3 & a_4 \\ a_1 & a_2 & a_3 & 1+a_4 \end{vmatrix} = 1 + a_1 + a_2 + a_3 + a_4$$ [2019]

11. Prove that-

$$\begin{vmatrix} -a & -b & c & d \\ b & -a & -d & c \\ c & -d & a & -b \\ d & c & b & a \end{vmatrix} = (a^2+b^2+c^2+d^2)^2$$ [2019]

12. Show that-

$$\begin{vmatrix} -1 & b & c & d \\ a & -1 & c & d \\ a & b & -1 & d \\ a & b & c & -1 \end{vmatrix} = (a+1)(b+1)(c+1)(d+1)\left[1-\frac{a}{a+1}-\frac{b}{b+1}-\frac{c}{c+1}-\frac{d}{d+1}\right]$$

[2018]

13. Evaluate -

$$\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}$$

14. Show that -

$$\begin{vmatrix} a+b+c & -c & -b \\ -c & a+b+c & -a \\ -b & -a & a+b+c \end{vmatrix} = 2(b+c)(c+a)(a+b)$$

15. Prove that -

$$\begin{vmatrix} a^2 & bc & ac+c^2 \\ a^2+ab & b^2 & ac \\ ab & b^2+bc & c^2 \end{vmatrix} = 4a^2b^2c^2$$

16. Prove that -

$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$$

17. Prove that -

$$\begin{vmatrix} -a & -b & c & d \\ b & -a & -d & c \\ c & -d & a & -b \\ d & c & b & a \end{vmatrix} = (a^2 + b^2 + c^2 + d^2)^2$$

18. Applying Laplace expansion prove that -

$$\begin{vmatrix} x+a & b & c & d \\ a & x+b & c & d \\ a & b & x+c & d \\ a & b & c & x+d \end{vmatrix} = x^3 (a + b + c + d + x)$$

19. By Applying Cramer's rule solve the following system of linear equations -

$$x + 2y - z = 1$$
$$3x - 2y + z = 0$$
$$2x + y + z = 2$$

20. Show that

$$\begin{vmatrix} 1+a_1 & a_2 & .... & a_n \\ a_1 & 1+a_1 & ... & a_n \\ a_1 & a_2 & .... & a_n \\ .... & .... & .... & .... \\ ... & .... & .... & ..... \\ a_1 & a_2 & .... & 1+a \end{vmatrix} = 1 + \sum_{i=1}^{n} a_i$$

Chapter-3
Matrix Algebra

1. Define–

- **i) Matrix** [2010,2017]
- **ii) Diagonal matrix** [2013]
- **iii) Skew-Symmetric matrix** [2010,2017]
- **iv) Idempotent matrix** [2019, 2014,2012,2016]
- **v) Nilpotent matrix** [2020, 2018,2012]
- **vi) Singular matrix** [2020, 2019, 2018,2015,2013, 2016]
- **vii) Adjoint matrix** [2011]

viii) Symmetric matrix [2019, 2014,2012,2010,2017,2016]

ix) Square matrix [2019, 2014,2012,2016]

x) Transpose matrix [2014,2011]

xi) Identity matrix [2015,2013,2011]

xii) Hermitian matrix [2020, 2018]

xiii) Inverse matrix [2018]

2. Define rank of the matrix. [2018]

3. If A and b are comparable matrix and A^T and B^T are the transpose of A and B respectively then prove that,

I) $(AB)^T = B^T A^T$ (II) $(A^T)^T = A$ [2019, 2014,2012,2017]

4. If A and B are non-singular matrices, then prove that,

(I) $(AB)^{-1} = B^{-1} A^{-1}$ (III) $(A^{-1})^{-1} = A$, and $(A^{-1})T = (A^T)^{-1}$ [2020, 2018,2013,2011,2016]

5. If A and B are idempotent matrices then AB is idempotent if AB = BA

6. If A and B are idempotent matrices, then A + B will be idempotent if and only if AB = BA = 0

7. If A= $\begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, prove that $A^3 = A^{-1}$. [2020]

8. Show that the matrix A = $\begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$ is idempotent. [2019]

9. If A = $\begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ then find $A^{-1}+2A^t$. [2018]

10. If A = $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{bmatrix}$

Find the values of $A^{-1}B$ and AB^{-1}. [2015]

Or,

If A = $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{bmatrix}$ and B = $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 5 \end{bmatrix}$

Find the values of A^{-1} B and AB^{-1}

11. If $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$ then find $\frac{1}{2}(A^{-1}+A)$. [2014]

12. Compute the value of x and y where $x + y = \begin{bmatrix} 7 & 0 \\ 2 & 5 \end{bmatrix}$ and $x - y = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$. [2013,2017]

13. If $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ then find A^{-1}. [2012]

14. If $B = \begin{bmatrix} 0 & 0 & 2 \\ 2 & 0 & 2 \\ 2 & 0 & 0 \end{bmatrix}$ find $B^{-1}B$. [2010]

15. If $A = \begin{bmatrix} 1 & 2 & 3 \\ -2 & 5 & -1 \\ 2 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 & 3 \\ 7 & -2 & 1 \\ 2 & 0 & -3 \end{bmatrix}$ then prove that $(AB)^T = B^T A^T$.

[2016]

16. Solve the following linear equation with the help of the matrix:-

$$x + 2y + z = 2$$
$$2x - y + 2z = -1$$
$$3x - 4y - 3z = -16$$

[2018]

17. Solve the following linear equations with the help of matrices-

$$3x + 5y - 7z = 13$$
$$4x + y - 12z = 6$$
$$2x + 9y - 3z = 20$$

[2013, 2017]

18. Solve the following system linear equations-

$$2x + 3y + 4z = 1$$
$$3x + 4y - 5z = 10$$
$$5x - 7y + 2z = 3$$

[2012]

19. Solve the following linear equation-

$$x + y + z = 6$$
$$x - y + z = 2$$
$$2x + y - z = 1$$

[2014]

20. Solve the following equations with the help of matrix-

$$x - 3y + 2z = 1$$
$$3x + 4y - 5z = 15$$
$$9x + 5y - 7x = 20$$

21. Solve the following system linear equations-

$$x + 2y + 3z + 4 = 0$$
$$2x + 4y + 5z + 7 = 0$$
$$3x + 5y + 6z + 10 = 0$$

[2019]

22. Find the rank of the matrix-

$$A = \begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -3 \\ 3 & 10 & -6 & -5 \end{bmatrix}$$

[2020, 2018]

Chapter – 4
Vectors In $\mathbb{R}^n$ and C^n

1. Define Vector in $\mathbb{R}^n$.
2. Define dot and cross product of two vectors in $\mathbb{R}^n$.
3. Define distance and norm in $\mathbb{R}^n$. [2019, 2018]
4. For any vectors u, v in R^n, then prove that, $|u \cdot v| \leq ||u||\ ||v||$
   or, state and prove cauchy – schwarz in equality.
5. For any vectors u, v $\in$c, prove that, $u.v = \overline{v.u}$
6. If $u = (u_1, u_2, \ldots\ldots, u_n)$ and $v = (v_1, v_2, \ldots\ldots, v_n)$ are vectors in R^n, then prove that $||u+v|| \leq ||u|| + ||v||$
   or, state and prove Minkkowski's inequality.
7. For any vectors $x, y, z \in \mathbb{R}^n$ prove that $(x + y)\cdot z = x\cdot z + y\cdot z$
8. For any complex numbers u, v $\in$ c prove that $\overline{u+v} = \overline{u} + \overline{v}$
9. Consider P(3, λ,–2) and Q(5,3,4) in IR^3. Find λ so that $\overrightarrow{PQ}$ is orthogonal to the vector (4,–3,2). [2018]
10. Consider the points, $P(3,\lambda,-2)$ and $Q(5,3,4)$ in IR^3. Find the value of λ so that $\overrightarrow{PQ}$ is orthogonal to the vector $(4,-3,2)$.. [2019]
11. Let u = (2, -1, 0, -3); v = (1, -1, -1, 3); w (1, 3, -2, 2) Find (i) 5u – 3v – 4w (ii) $u \cdot v$ (iii) ||w||
12. Let u = (4, 1, 2, 3) and v = (0, 3, 8, - 2) then evaluate the following expressions:-
    (i) $u \cdot v$ (ii) ||u + v||
13. Let u (1, 2, 3, -4); v = (5, -6, 7, 8), find (i) u + v (ii) ||u|| (iii) d (uv) (iv) $u \cdot v$

14. Suppose u = (2 + 3i, 4 − i, 1 − 2i) and v = (3 − 2i, 5i, 4 − 6i) Find $u \cdot v$, ||u|| and (2 − 3i) v

15. Find the dot products u.v and v.u where u (3 − 2i, 5i, 1 + 3i) and v = (5 + 2i, 2 − 3i, 7 + 4i)

Chapter - 5

Vector Spaces

1. Define vector space (or linear space).
2. Define subspace of a vector Space.
3. Define linear combination of vectors.
4. Define linear span of a subset of a vector space.
5. Define linear dependence and linear independence.
6. Define basis and dimension of a vector space.
7. Define coordinates of a vector relative to a basis.
8. The intersection of two subspace S and T of a vector space v is also a subspace of V.
9. Let S be a non-empty subset of a vector space V, then L (S) is a subspace of V containing S. Furthermore, if W is any other subspace of V containing S, then L (S)$\subset$ w.
10. The vector space V is the direct sum of its subspaces S and T if and only if (i) V = S + T and (ii) S $\cap$ T = {0}.
11. The non-zero vectors $v_1, v_2, \ldots\ldots, v_n$ in a vector space V are linearly dependent if and only if one of the vectors V_k is a linear combination of the preceding vectors $v_1, v_2, \ldots\ldots, v_{k-1}$.
12. If S and T are subspaces of a finite dimensional vector space V over the field F then prove that dim (S + T) = dim S + dim T − dim (S $\cap$ T).
13. Show that T = {(a, b, c, d)} $\in \mathbb{R}^4$: 2a − 3b + 3c − d} is a subspace of $\mathbb{R}^4$.
14. Show that w = {a, b, c) | a, b, c $\in \mathbb{R}$ and 2a − b + c = 10} is not a subspace of $V_3(\mathbb{R})$.
15. Consider the vectors v_1 = (2, 1, 4), v_2 = (1, −1, 3) and v_3 = (3, 2, 5) in IR^3. Show that v = (5, 9, 5) is a linear combination of v_1, v_2, v_3.
16. Whether or not the vectors (1, 2, 6) is a linear combination of the vectors $u_1 = (2, 1, 0)$, $u_2 = (1, -1, 2)$ *and* $u_3 = (0, 3, -4)$.
17. In the vector space IR^3 express the V = (1, −2, 5) as a linear combination of the vectors v_1 = (1, 1, 1), v_2 = (1, 2, 3) and v_3 = (2, −1, 1).
18. Express the matrix A = $\begin{bmatrix} 5 & 1 \\ -2 & 3 \end{bmatrix}$ as a linear combination of the matrices

$A_1 = \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$, $A_2 = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ and $A_3 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

19. Show that the set of vectors {(3, 0, 1, −1), (2, −1, 0, 1), (1, 1, 1, −2)} is linearly dependent.

20. let u, v and w are independent vectors, show that u + v, u − v, u − 2v + w are also independent.

21. Test the dependency of the following sets :

i) {(1, 2, −3), (2, 0, −1), (7, 6, −11)}

ii) {(2, 0, −1), (1, 1, 0), (0, −1, 1)}

22. let p(t) the vector space of all polynomials of degree ≤ 3 over the real field $\mathbb{R}$. Determine whether the following polynomials in P(t) are linearly dependent or independent :

$u = t^3 + 4t^2 - 2t + 3$, $v = t^3 + 6t^2 - t + 4$ and $w = 3t^3 + 8t^2 - 8t + 7$

23. Let S and T be the following subspaces of $\mathbb{R}^4$:

$S = \{(x, y, z, t) \mid y - 2z + t = 0\}$

$T = \{(x, y, z, t) \mid x - t = 0, y - 2z = 0\}$

Find a basis and the dimension of i) s ii) T iii) $S \cap T$

24. let S and T be the following subspaces of $\mathbb{R}^4$:

$S = \{(x, y, z, t) \mid x + z + t = 0\}$

$T = \{(x, y, z, t) \mid x + y = 0, z = 2t\}$

25. Find the dimension and a basis of the solution space w of the following system

$$\begin{aligned} x + 2y + 2z - s + 3t &= 0 \\ x + 2y + 3z + s + t &= 0 \\ 3x + 6y + 8z + s + 5t &= 0 \end{aligned}$$

26. Giving the vectors (2, 1, 1), (1, 3, 2), (1, 3, -1) and (1, -2, 3). Test whether they are linearly independent by sweep out method.

27. Show that the set S = {(1, 0, 0), (1, 1, 0), (1, 1, 1)} is a basis of $\mathbb{R}^3$ and hence find the coordinates of the vector (a, b, c) with respect to the above basis.

Chapter-6
Linear Transformation

1. Define linear transformation. [2020, 2019, 2018]
2. Define kernel and range of linear transformation. [2019]
3. Define Isomorphism.
4. Define singular and non-singular linear transformations.
5. Show that the product of two linear transformations is a linear transformation. [2019]
6. Let u and v be two vector space s over the same field F and T_1 and T_2 be linear transformations from u into v. Then T_1+T_2 is a linear transformations. [2018]
7. If $T : U \to V$ is a linear transformation, then

   i) the kernel of T is a subset of U,

   ii) the range of T is a subset of V.
8. A linear transformation $T : U \to V$ is an isomorphism if and only if it is non-singular.

9. Show that the mapping $T: IR^2 \to IR^3$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_1)$ is linear. [2018]

10. Show that the transformation $T: R^2 - R^2$ defined by $T(x,y) = (x + 2y, 2x - y)$ is linear transformation on R^2.

11. Show that the transformation defined by $T(x,y,z) = (x + 2y, y - z, x + 2z)$ is a linear transformation.

12. Let S and T be the liner transformation S, T: $IR^3 \to IR^3$ defined by $T(a,b,c) = (a,b,0) \forall a,b,c \in \mathbb{R}$. Show that T is a linear transformation.

[2020]

13. In $\mathbb{R}^2$ consider the function T, $\mathbb{R}^2 - \mathbb{R}^2$ given by $T(x, y) = (x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta)$ where θ is a fixed number and $0 \leq \theta \leq 2\pi$. Then T is a linear transformation.

14. Let S and T be the liner operators of $\mathbb{R}^2$ into $\mathbb{R}^2$ defined by $S(x,y) = (3x + 2y, -6x + y)$, $T(x,y) = (2x + y, x - y)$. Find S + T, ST, TS, and S^2. [2019]

15. Let S and T be the liner transformation S, T: $IR^3 \to IR^3$ defined by $S(x, y, z) = (y, z, x)$ and $T(x, y, z) = (x+y+z, 0, 0)$. Find -

i) (TS) (1, 0, 1)

ii) (ST) (1, 0, 1)

iii) (S+T) (1,0,1)

iv) (S-T) (1,0,1) [2018]

16. Show that the linear operator $T: \mathbb{R}^3 \to \mathbb{R}^3$ defined by $T(x, y, z) = (x + y - z, x - y + z, x + y + z)$ is invertible or not. If the operator is invertible, find T^{-1} and also verify the result.

17. Show that each of the following linear operators T on $\mathbb{R}^3$ is invertible and find a formula for T^{-1}.

18. let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear transformation defined by $T(x, y, z, t) = (x - y + z + t, x + 2z - t, x + y + 3z - 3t)$. Find a basis and dimension of (i) Range space of T (ii) null space of T.

19. Let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x, y, z) = (x + 2y, y - z, x + 2z)$ Find the rank and nullity of T.

Or, let $T: \mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by $T(x, y, z) = (x + 2y, y - z, x + 2z)$. Find (i) Im T and (ii) Ker T.

20. $T: \mathbb{R}^3 \to \mathbb{R}^3$ be a linear operator defined by $T(x,y,z) = (x + 2y - 3z, 2x - y + 4z, 4x + 3y - 2z)$. Find a basis and the dimension of the (i) Im T of T (ii) Ker T of T. [2020]

21. Find a linear transformation $T: \mathbb{R}^3 \to \mathbb{R}^4$ whose ImT is generated by {(1, 2, 0, -4), (2, 0, -1, -3)}. [2019]

22. Consider the matrix mapping

$A = \begin{bmatrix} 1 & 3 & 4 \\ 3 & 8 & 10 \\ -2 & -3 & -2 \end{bmatrix}$ **Find a basis and the dimension of (i) the image of A and (ii) the Kernel of A.**

Topic: 7
Matrix Representation

1. Define matrix representation of the linear transformation. **[2020]**

2. Let $\{u_1, u_2, \ldots\ldots, u_m\}$ be a basis of U and T be any linear operator of U, then for any vector $W \in U$ show that $[T]_u[w]_u = [T(w)]_u$. **[2020]**

3. Let T : $\mathbb{R}^3 \to \mathbb{R}^3$ be the linear mapping defined by T (x, y, z) = (2x + y, x − y, z) :Find the matrix representation of T relative to the usual basis of $\mathbb{R}^3$.

4. Let T : $\mathbb{R}^3 \to \mathbb{R}^3$ be the linear operator defined by T(x, y, z) = (x − y, y − x, x − z). Find the matrix of T with respect to the basis U= $\{u_1, u_2, u_3\}$ where $u_1 = (1, 0, 1), u_2 = (0, 1, 1)$ *and* $u_3 = (1, 1, 0)$.

5. Let T : $V_3(\mathbb{R}) \to V_2(\mathbb{R})$ be he linear transformation defined by T (x, y, z) = (2x+ y − z, 3x − 2y + 4z). Find the matrix of T relative to the basis {(1, 1, 1), (1, 1, 0), (1, 0, 0)} of $V_3(\mathbb{R})$ and the basis {(1, 3), (1, 4)} of $V_2(\mathbb{R})$.

6. let T : $\mathbb{R}^2 \to \mathbb{R}^2$ be the linear operator defined by T (x, y) = (x + y, −2x + 4y). Find the matrix of T respect to the basis $u = (u_1, u_2)$ when $u_1 = (1, 1)$ and $u_2 = (1, 2)$.

Chapter-8
Eigenvalues and Eigenvectors

1. Define eigenvalues and eigenvectors. **[2020, 2018]**

2. Define characteristic polynomial Matrix, and characteristic equation. **[2019]**

3. State and prove Cayley-Hamilton Theorem. **[2020, 2019]**

4. Find the eigen values and the corresponding eigen vectors of the matrix-

$$\begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$$ **[2018]**

5. Find the eigenvalues and eigenvectors of the matrix-

$$\begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

6. Find the characteristic values, the characteristic vectors, and a diagonalzing matrix P for the matrix

$$A = \begin{bmatrix} 2 & -2 & 3 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$$

7. Find the characteristic equation of the matrix-

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ 3 & 1 & 1 \end{bmatrix}$$ **and verify Cayley-Hamilton theorem for it.**

8. Using Cayley-Hamilton theorem find the inverse matrix $A = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ **[2019]**

9. Using Cayley-Hamilton theorem find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$

10. Let $A = \begin{bmatrix} 2 & 2 \\ 3 & -1 \end{bmatrix}$ **find** $g(x) = x^2 - x - 8$.